University of Warwick

Course Code: IB94V0

Course Name: Data Analytics & Artificial Intelligence

Individual Assignment

Purchase Decision Pattern Analysis in E-Commerce

1. Business Problem

Problem

- E-commerce platforms trigger marketing interventions based on add-to-cart events(Jiang and Turut, 2024; Wang et al., 2023), and static customer labels(Indiani et al., 2024; Vallarino, 2023), yet purchase decisions vary systematically in search depth, duration, and experience(Necula, 2023; Vallarino, 2023), making uniform intervention mismatched to actual decision needs(Necula, 2023).

Significance

- Mismatched interventions generate two distinct costs: disrupting high-experience decisions that are already efficient, and failing to address the sources of high-friction decisions — both reduce conversion quality and waste marketing resources(McKinsey, 2013).

Pain Point

- Online carts are frequently used as organisation tools rather than signals of purchase intent(Kukar-Kinney and Close, 2010), meaning add-to-cart events could misrepresent decision readiness — making trigger-based interventions mistimed across heterogeneous decision types.

Stakeholders

- E-commerce operators for intervention insights; product managers for purchase funnel UX optimisation; data analysts for decision-state classifier development.

2. Dataset

Selection

- Retailrocket: 2,756,101 clickstream events from a live e-commerce site, May–Sep 2015.

Suitability

- Millisecond-precision timestamps enable direct construction of decision duration and search depth as objective variables — a methodological advantage over self-report measures.

Key Variables

Variable	Definition	Role
decision_hours	First view to first transaction (hours)	Outcome: decision friction
research_views	View count before purchase	Outcome: search depth
has_addtocart	Add-to-cart before purchase (0/1)	Behavioural signal
total_purchases	Purchases completed before this decision	Predictor: prior experience
active_days	Days since visitor's first platform event	Predictor: platform familiarity

Scale

- Final sample: 18,738 decision units from 11,719 visitors.

3. Data Pre-processing

Cleaning

- Timestamps converted to datetime; analysis unit defined as visitorid+itemid pairs (decision-level); excluded 2,532 records where research_views=0 (no observable platform decision path).

Missing Values & Outliers

- Five right-skewed variables (max skew 15.97) were log-transformed for clustering and regression; raw values retained for descriptive statistics and decision tree. view_interval_mean excluded from main analysis due to 54.6% structural zeros; retained for sub-sample analysis.

Feature Engineering

- **total_purchases** and **active_days** were constructed as cumulative-to-decision-timestamp values to avoid look-ahead bias; using full visitor history would underestimate the **log_total_purchases** regression coefficient from −0.21 to −0.14.

- Price was inferred from item_properties.csv property 790 (time-varying numeric field, 91.3% match rate); excluded from clustering as a product attribute; used for post-hoc profiling.

External Data

- No external data. item_properties.csv is part of the same package.

4. Analysis

Methods

- K-Means clustering (k=4, silhouette=0.500) on five log-transformed behavioural variables; OLS regression (with/without log_price) benchmarked against Ridge, Lasso, and Random Forest via 5-fold cross-validation; decision tree (test accuracy=91.3%) to translate cluster labels into business-readable rules; sub-sample Spearman correlation and Kruskal-Wallis test on view_interval_cv vs decision_hours (n=4,055).

Rationale

- K-Means requires no pre-specified outcome variable, suited to exploratory pattern; silhouette score validated cluster separability. OLS preferred over Random Forest (R^2=0.37 vs 0.54) for interpretable effects supporting intervention design. Decision tree depth=3 selected, balancing accuracy against rule readability.

Analytical Process

- Descriptive statistics → log1p transformation → correlation matrix and collinearity check → OLS regression → quadrant EDA → K-Means → Kruskal-Wallis cluster validation → decision tree → sub-sample browsing regularity analysis.

5. Results and Interpretation

Key Findings

- Four decision clusters are statistically distinct (KW H=7,228, p≈0):

Cluster	Size	Decision time (h)	Prior purchases	Active (days)
High-experience	24.2%	0.07	39 (median)	18

Lightweight	54.7%	0.18	0	0.02
High-friction	12.4%	65	0	7
No-addtocart	8.7%	0.11	0	0.02

- OLS regression (R^2=0.37, n=17,107):

Variable	Coefficient	p-value
log_research_views	+1.69	<0.001
log_total_purchases	−0.21	<0.001
log_active_days	+0.42	<0.001
has_addtocart	n.s.	0.88 (Lasso coeff.=0)

- Sub-sample (research_views≥3, n=4,055): decisions with irregular inter-view gaps (high view_interval_cv) have a median decision time of 21.4h versus 0.43h for regular browsers (KW H=1,285, p≈0; Spearman ρ=0.61).

Interpretation

- The four clusters reflect structurally different decision mechanisms: high-friction and lightweight decisions share zero prior purchases, yet differ by 7 days of active browsing and 65 hours of decision time.
- total_purchases and active_days capture distinct constructs: experience reduces cognitive load; active_days conflates platform familiarity with prolonged deliberation and cannot substitute for it.
- Irregular inter-view gaps represent interrupted return visits consistent with perceived-risk hesitation, not ongoing information search.

Business Meaning

- Platforms using add-to-cart as trigger could both mistime and mistype interventions: the signal predicts neither decision speed (p=0.88) nor reaches all purchases, while the high-friction segment which most in need of intervention, is distinguishable through active_days and browsing irregularity.
- Using active_days as an experience proxy compounds the problem: it would classify high-friction decisions as their opposite, experienced users not needing intervention.

- Irregular browsing cadence provides a platform-observable signal for identifying confidence-deficit decisions before they abandon, enabling proactive delivery rather than reactive recovery.

6. Business Insights and Recommendations

Insights

- H1 — Experience accelerates decisions; activity time does not: high-experience decisions complete in 0.07h even at the highest median price (59,880 vs 45,000 for lightweight). Platforms that use active_days as an experience proxy will systematically under-intervene on their highest-friction segment.
- H2 — High-friction decisions lack confidence, not information: Cluster 2 had already browsed 4 times across 7 days before purchasing; irregular inter-view gaps predict 50× longer decisions, consistent with perceived-risk hesitation rather than information search.
- Supplementary — Cluster 1 (54.7%) is the low-intervention-value majority: decisions are fast, frictionless, and require only path efficiency. Cluster 3 (8.7%) represents a unreachable segment: no-addtocart decisions complete within minutes, placing them outside cart-triggered intervention reach and quantifying an ~9% coverage ceiling.

Recommendations

- High-experience decisions: streamline the purchase path (one-click reorder, reduced confirmation steps); avoid content that disrupts an efficient process.
- High-friction decisions: deploy trust-reducing content (verified reviews, return guarantees) rather than product information; use irregular return-visit cadence as an early signal to pre-load trust content.
- Platform-wide: acknowledge the ~9% structural blind spot in cart-triggered strategies; investigate earlier behavioural signals (repeated views) as supplementary triggers.

Limitations

- Sample restricted to completed purchases only; non-converting visitors are unobservable, limiting generalisation to the full conversion funnel.
- All findings are associational; the confidence-deficit interpretation of Cluster 2 is theoretically motivated but not causally identified.

- Price is inferred from an unlabelled item property; its negative regression coefficient (higher-price items associated with faster decisions) remains an open interpretive question.

Future Directions

- A/B experiments to test trust-content interventions on high-friction decisions.
- Survival analysis to incorporate non-converting visitors as right-censored observations.
- Product category analysis using categoryid to control for item-type.

References

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